

A 56-Mc. Transmitter for Mobile Work

Simple Construction of a 12-Watt A.M. Rig

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Although the design of a mobile 56-Mc. transmitter must usually be tailored to meet the restrictions set by the type of automobile used, here is a transmitter that can be used in practically any car that has a trunk rack. A dash-board control system is also described, with suggestions for installing the power supply.

THE design of a fixed-station transmitter usually starts with the final stage and works back through the exciter stages to the power supply, but the design of a portable or mobile rig hinges on the available power supply. Batteries can be used for portable or mobile rigs, but their use is generally limited to the low-power class of not more than one or two watts input. The next group of available power supplies falls around the 30- to 40-watt class and includes vibrator packs and small motor-generators. Above that one jumps into a power class where the design is not dictated quite as much by the power supply as by how much one can afford to pay.

The transmitter to be described was designed to work from a power supply delivering 125 ma. at 325 volts. Working from a 300-volt supply brings up the current drain slightly above rating and is not recommended except in case of emergency. This puts it in the class served by vibrator packs and small motor-generators that are not expensive and can be operated from the regular automobile battery without overload. Since maximum economy is desired in the exciter and audio stages, high-gain doubler tubes and Class-B audio for modulation were used. A minimum excitation requirement dictates the use of a beam tube as the modulated amplifier.

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The Circuit

After trying several experimental arrangements, the line-up shown in Fig. 1 was selected as a very practical one. A 6AG7 Tri-tet oscillator using a 7-Mc. crystal — considered to be more sure-fire than a 14-Mc. crystal — quadruples in its plate circuit to 28 Mc. to drive a 6AG7 doubler to 56 Mc. The 6AG7 is a high-gain pentode designed for television work, and it makes an excellent crystal oscillator or frequency multiplier for amateur work. The output of the doubler is used to drive a 6V6 amplifier on 56 Mc. A 6L6 was tried instead of the 6V6 but showed no improvement in any way at the input the amplifier runs (12 watts). Provision for neutralizing the 6V6 was included at first, but it was found unnecessary in this particular parts arrangement. It is not to be assumed, however, that the 6V6 will work well at 56 Mc. without neutralization in every arrangement — the necessary neutralizing capacity is small and is doubtless present in this layout as a stray capacity. The grid of the 6V6 is tapped down on the driver plate coil to lighten the loading.

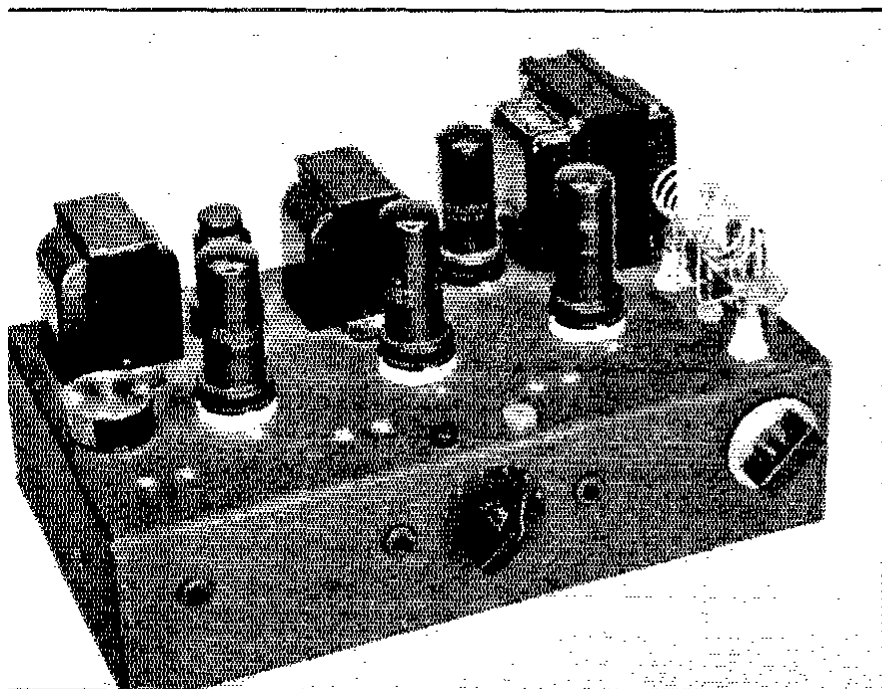
The modulation equipment consists of a 6C5 driver stage and a 6N7 Class-B modulator. Anything except a single-button microphone would have required more audio gain and would have introduced the possibility of more hash pick-up because of the lower level of the output of the microphone. While single-button microphones do not give "broadcast quality," there can be little or no complaint about their effectiveness and reliability, important factors in any portable/mobile unit.

Construction

The transmitter is built on a 7- by 12- by 3-inch chassis, thus providing plenty of room for the parts. Reference to the photographs will show the placement of parts, but some of the minor

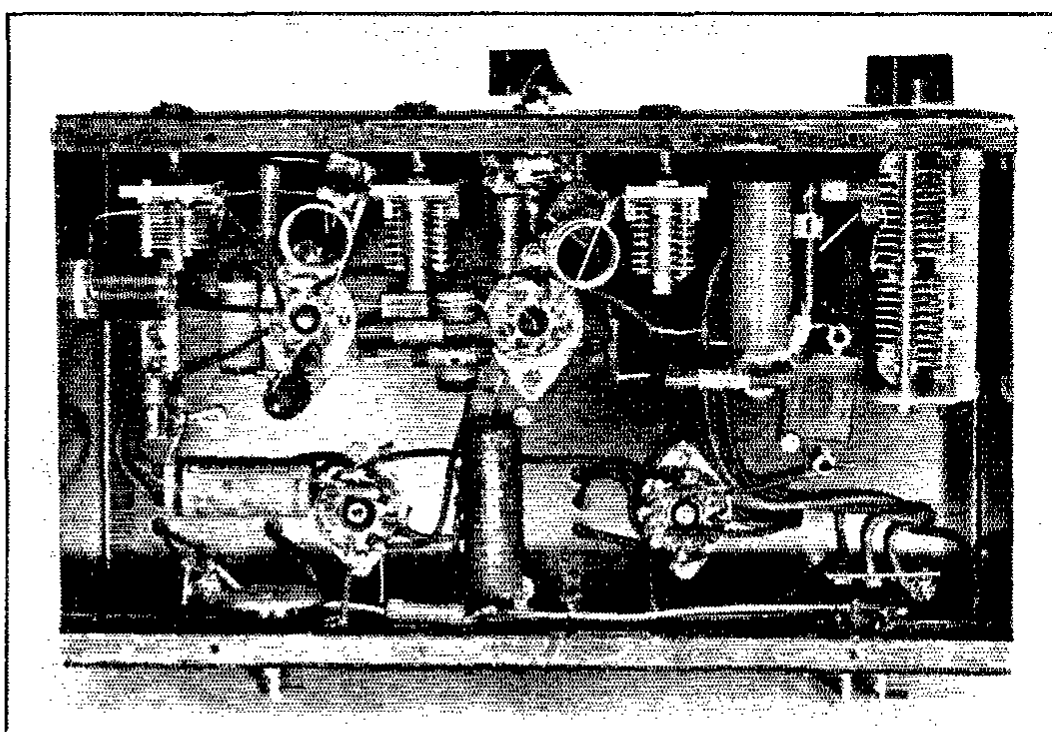
A complete 12-watt 56-Mc. 'phone transmitter, ready for installation in car or home. The tubes along the front, from left to right, are 6AG7 Tri-tet oscillator, 6AG7 doubler and 6V6 final amplifier. The 6C5 driver (left) and the 6N7 Class-B modulator are at the rear between the transformers. The knob on the right controls the final tank condenser — the other tuning condensers are adjusted by screw driver through the rubber grommets. The meter switch is mounted on the front center, just under the meter pin jacks.

Note that the antenna coil is mounted on the antenna binding post strip — coupling is adjusted by swinging the coil.



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A view under the chassis of the 56-Mc. transmitter shows the straightforward arrangement of parts. The coils L_2 and L_3 are self-supporting and are mounted on their respective condensers. Note the audio volume control and the power supply plug mounted at the rear of the chassis. The microphone lead from the plug to the microphone transformer is run through grounded shield braid.



constructional points should be pointed out. The tuning condensers C_1 , C_2 and C_3 are mounted on the underside of the chassis on the small brackets that are furnished with them, and they are set far enough back from the front so that the ends of the shafts do not quite touch the metal. They are adjusted by a screw driver that

is prevented from shorting to the chassis by

rubber grommets in the holes. The final tank condenser, C_4 , is supported on the panel.

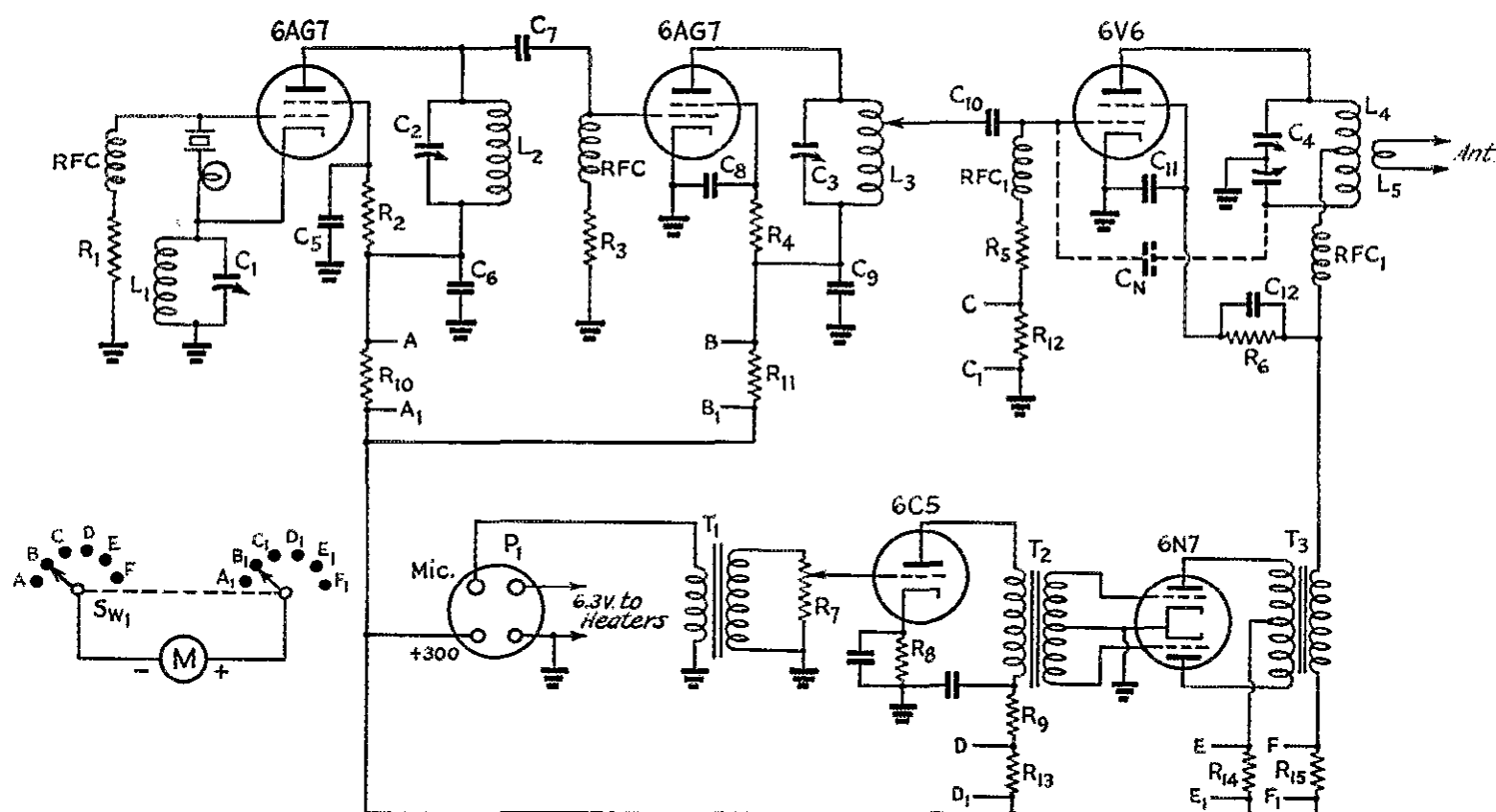


Fig. 1 — Wiring diagram of the 56-Mc. 'phone transmitter.

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| C_1 — 50- μ fd. variable (National UM-50). | C_{13} — 25- μ fd., 25-volt electrolytic. | RFC — 2.5-mh. r.f. choke (National R-100U). |
| C_2 , C_3 — 25- μ fd. variable (National UMA-25). | CN — See text. | RFC ₁ — U.h.f. r.f. choke (Ohmite Z1). |
| C_4 — 30- μ fd. per section variable (Hammarlund HFD-30-X). | R_1 , R_3 — 0.2 megohms, 1-watt. | Sw ₁ — 2-circuit, 5-position rotary switch, non-shorting (Mal-lory 3226J). |
| C_5 , C_8 — 0.01- μ fd., 400-volt paper. | R_2 , R_4 — 40,000 ohms, 1-watt. | T_1 — Microphone transformer (Stancor A-4726). |
| C_6 , C_9 , C_{11} — 0.002- μ fd. mica. | R_5 — 30,000 ohms, 1-watt. | T_2 — Driver transformer (Stancor A-4721). |
| C_7 , C_{10} — 250- μ fd. mica. | R_6 — 5,000 ohms, 2-watt. | T_3 — Modulation transformer (Stancor A-3845). |
| C_{12} , C_{14} — 8- μ fd., 450-volt electrolytic. | R_7 — 0.1-megohm volume control. | L_1 — 19 turns No. 18 enam., spaced slightly to occupy $\frac{3}{8}$ -inch winding length, on $\frac{3}{4}$ -inch diam. form (National PRF-2). |
| T_3 — Modulation transformer (Stancor A-3845). | R_8 — 1000 ohms, $\frac{1}{2}$ -watt. | L_2 — 8 turns No. 14, spaced to occupy $1\frac{7}{8}$ inch, $\frac{7}{8}$ -inch diam., self-supporting. |
| L_1 — 19 turns No. 18 enam., spaced slightly to occupy $\frac{3}{8}$ -inch winding length, on $\frac{3}{4}$ -inch diam. form (National PRF-2). | R_9 — 6000 ohms, 1-watt. | L_3 — $3\frac{1}{2}$ turns No. 14, spaced to occupy $\frac{7}{8}$ inch, |
| L_2 — 8 turns No. 14, spaced to occupy $1\frac{7}{8}$ inch, $\frac{7}{8}$ -inch diam., self-supporting. | R_{10} – R_{15} — 25 ohms, $\frac{1}{2}$ -watt. | L_4 — 3 turns No. 14, each side center-spaced to occupy $\frac{3}{4}$ inch, $\frac{7}{8}$ -inch diam. |
| L_3 — $3\frac{1}{2}$ turns No. 14, spaced to occupy $\frac{7}{8}$ inch, | | L_5 — 2 turns No. 14, $\frac{7}{8}$ -inch diam. |
| | | P_1 — 4-prong base-mounting plug (Amphenol RCP-4). |
| | | Lamp in series with crystal is 60-ma. dial light. |

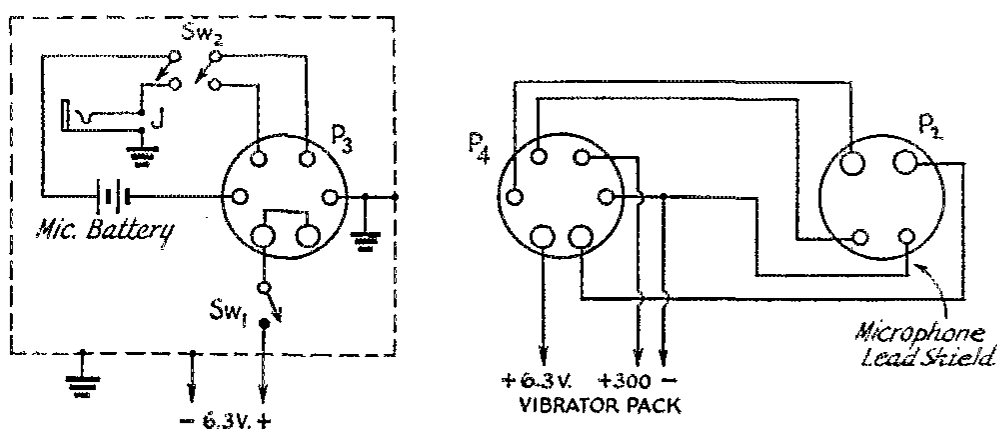


Fig. 2 — Circuit diagram of control box.

J — Small microphone jack (Mallory 702B).
 Sw₁ — D.p.s.t. high-current toggle with sections in parallel.
 Sw₂ — D.p.s.t. toggle.
 P₂ — 4-prong cable socket (Amphenol PF-4).
 P₃ — 6-prong cable plug (Amphenol RCP-6).
 P₄ — 6-prong socket (Amphenol PF-6).
 Battery is Burgess 3A2. Microphone lead is shielded throughout.

All of the inductances are mounted on or near their respective tuning condensers except the final tank coil, L_4 , which is mounted above the chassis on feed-through insulators. This makes it more convenient to adjust the antenna coupling coil, L_5 , after installing the transmitter in the car.

The plate circuits and the final grid circuit can be metered by plugging in the meter leads to the two pin jacks on the front center of the chassis and setting the meter switch to the proper position. This is a convenience when tuning up with a different crystal or antenna. The power leads are terminated at a four-prong plug mounted on the back of the chassis.

Control Circuit

One problem in connection with mobile units is the drop in the line from the battery to the vibrator or motor-generator unit, and these leads must be kept as short as possible. This transmitter is intended to be mounted in the trunk rack of the car, with the control box mounted on the dashboard of the car and the vibrator pack mounted under the hood on the fire wall. This is, of course, for a car with the battery under the hood — for cars with the battery elsewhere the vibrator pack and control box might have to be mounted differently. There isn't too much drop in the leads running back to the heaters of the tubes from the battery if heavy wire is used, and the drop in the 300-volt line from the vibrator pack is negligible.

The wiring diagram of the control box is shown in Fig. 2. As can be seen, the microphone battery is mounted in this box, and a jack is provided for the microphone. The switch Sw_1 turns on the vibrator pack and the heaters of the tubes, while switch Sw_2 is used as an "on-off" switch for the transmitter, since it controls the microphone battery and the plate supply lead. The control box is a small 4- by 4- by 2-inch box (Parmet MC-442) and takes up very little room.

An alternative system is to mount the vibrator pack and an additional storage battery in the trunk rack and to control both the "on-off" of the heaters and vibrator pack and of the plate power through suitable relays controlled from the dash. However, the storage battery must be removed from the car for charging, and thus the installation may not be always "ready to go."

Tuning the Transmitter

The adjustment of the transmitter is conventional in every way and it should not be necessary to repeat here how to tune a crystal-controlled transmitter. With 325 volts from the power supply, the total plate and screen currents of the 6AG7 Tri-tet and the 6AG7 doubler will be 12 and 16 ma. respectively, and the final grid current should run about 2 ma. If, when the voltage is removed from the screen and plate of the 6V6 final, there is no flicker in the grid current as the final tank is tuned through resonance, there is no need to worry about neutralizing the final amplifier. However, if a flicker (of 0.1 ma. or so) does show up, the amplifier can be neutralized readily by running a stiff wire from the free end of the final tank over near the grid terminal on the 6V6 socket to form a neutralizing condenser (shown by dotted lines in Fig. 1). The stage is then neutralized in the usual manner, varying the neutralizing capacity by moving the free end of the wire. Connecting the voltage to the screen and plate of the 6V6 and tuning to resonance, the total plate and screen current should run under 35 ma. unloaded and about 39 or 40 ma. loaded.

The 6C5 plate current will run around 8 ma., the no-signal 6N7 plate current around 35 ma., kicking up to about 50 ma. on peaks.

The antenna can be anything from 0.25- to 0.6-wavelength long, depending upon what one has available and what the XYL's aesthetic sense will stand. Since the transmitter can be mounted close to the end of the antenna, there is no particular problem in feeding the antenna aside from finding a suitable insulator to run through the side of the car. If something near a quarter-wavelength long is used for the antenna, one side of the antenna coil, L_5 , should be grounded to the car and a variable condenser connected in series with the antenna and the other side of L_5 . When the antenna is near a half-wavelength long, parallel tuning of L_5 should be used. The center of L_5 can be grounded or the whole thing can be left floating. Regardless of the length of antenna, the antenna coupling is varied by movement of L_5 with respect to L_4 after tuning both amplifier and tank circuit to resonance.